

# OECD Data Quality and Validation

From raw national-accounts data to structured, tested outputs  
Internship & Thesis Project

R

Excel

Data Cleaning

Quality Assurance

Economic Analysis

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## Project overview

|       |                               |       |                                |
|-------|-------------------------------|-------|--------------------------------|
| Role  | Economic Data Analysis Intern | Data  | OECD national accounts, Italy  |
| Tools | R, RStudio and Excel          | Scope | Annual and quarterly workflows |

## The challenge

Official OECD data are detailed and reliable, but they are not immediately ready for analysis. The source uses long labels, multiple datasets, institutional-sector codes and different annual and quarterly structures. The project addressed the gap between raw statistical data and clear, consistent analytical outputs.

## Project scope

### 1. Non-financial accounts

Mapped OECD transactions into compact, sector-specific variables and created QA checks.

### 2. Financial architecture

Adapted the mapping logic to instruments, assets/liabilities and flows/stocks.

### 3. Transaction-flow matrix

Reorganized sector data into signed flows and tested accounting consistency.

Portfolio focus: This case study focuses on the non-financial accounts and transaction-flow matrix stages of the project.

## From raw data to tested outputs

The work combined economic interpretation with repeatable data-processing and validation steps.



## What I worked on

- Organized annual and quarterly OECD datasets by institutional sector.
- Used R and Excel to clean, relabel and structure economic variables.
- Investigated missing values, duplicate representations and label collisions.
- Applied checks for mapping coverage, duplicates and unused classifications.
- Supported construction of signed transaction-flow matrices.
- Documented decisions, limitations and validation results.

## Three decisions that improved data quality

|                              |                                                                                                                                                                                        |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Annual vs quarterly</b>   | Separate scripts were retained because the OECD datasets differ structurally. Shared naming and QA principles kept the outputs consistent without hiding source differences.           |
| <b>Missing is not zero</b>   | Unavailable observations, structural zeros and derived values were kept distinct. This prevented incomplete data from appearing artificially balanced.                                 |
| <b>Residuals as evidence</b> | A proposed household adjustment was removed after testing traced the discrepancy to duplicate saving representations. Remaining small review rows were retained rather than concealed. |

Why this matters: Reliable analysis depends on knowing which values are observed, derived, missing or only mechanically balanced. The project treated validation as part of the analysis, not as a final cosmetic check.

## What the project demonstrates

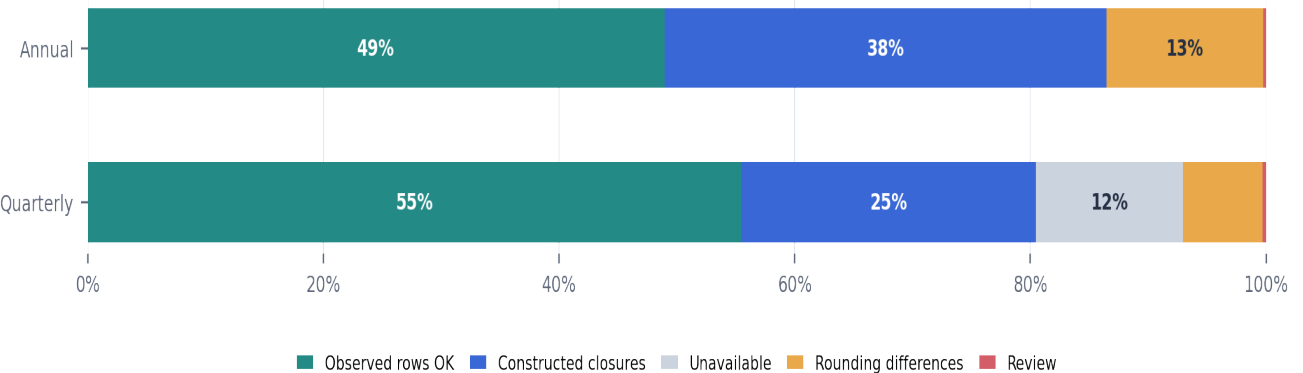
|                                                                                                               |                                                                                                                                      |                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Data quality</b><br>Cleaning, classification, missing-data handling, duplicate detection and traceable QA. | <b>Analytical reasoning</b><br>Testing discrepancies, separating source limitations from coding problems and preserving uncertainty. | <b>Communication</b><br>Documenting methods, assumptions and results so another person can review the workflow. |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|

## Evidence and outcomes

The final workbooks include sector tables, complete time-series matrices, long-form outputs, row-closure checks, net-lending validation, Total Economy comparisons and input-availability records.

|                                        |                                                   |                                                     |                                                        |
|----------------------------------------|---------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|
| <b>16</b><br>annual years<br>2010-2025 | <b>65</b><br>quarterly periods<br>2010 Q1-2026 Q1 | <b>€0.2m</b><br>maximum annual<br>domestic NL error | <b>€0.3m</b><br>maximum quarterly<br>domestic NL error |
|----------------------------------------|---------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|

### Row-closure validation outcomes



Annual: 188 observed rows OK, 144 constructed closures, 51 rounding differences and 1 review row. Quarterly: 865 observed rows OK, 390 constructed closures, 195 unavailable production closures, 105 rounding differences and 5 review rows.

**Example output: selected 2025 transaction flows**

| Selected 2025 flow                                      | NFC     | FC     | Government | Households | ROW   |
|---------------------------------------------------------|---------|--------|------------|------------|-------|
| P1 Output                                               | +3128.7 | +168.8 | +411.6     | +700.9     | +0.0  |
| P2 Intermediate consumption                             | -2033.8 | -71.3  | -133.0     | -157.4     | +0.0  |
| P3 Final consumption expenditure                        | +0.0    | +0.0   | -412.1     | -1299.9    | +0.0  |
| P5 Gross capital formation                              | -279.1  | -9.0   | -86.9      | -127.4     | +0.0  |
| D1 Compensation of employees                            | -608.2  | -38.6  | -203.8     | +856.8     | -6.1  |
| D5 Current taxes on income, wealth, etc.                | -47.7   | -13.5  | +344.1     | -282.6     | -0.3  |
| D61 Net social contributions                            | +22.7   | +20.2  | +305.9     | -349.2     | +0.4  |
| D62 Social benefits other than social transfers in kind | -22.1   | -14.7  | -459.2     | +499.5     | -3.4  |
| D41 Interest                                            | -8.5    | +36.7  | -81.5      | +26.6      | +26.7 |
| D9 Capital transfers                                    | +25.7   | +1.3   | -29.7      | +8.7       | -6.0  |

Values in € billions. Receipts are positive and payments are negative. The full workbook retains all rows, periods and validation fields.

**Project materials produced****R workflows**

Annual and quarterly scripts for mapping, cleaning, structuring and TFM construction.

**Validation workbooks**

Row-closure, net-lending, Total Economy and input-availability checks.

**Documentation**

Recorded assumptions, decisions, limitations and review results.

Project takeaway: I treated data validation as part of the analysis itself: understand the source, make assumptions explicit, build repeatable transformations and test the result before presenting it.

|                    |              |                        |                              |                      |
|--------------------|--------------|------------------------|------------------------------|----------------------|
| <b>R / RStudio</b> | <b>Excel</b> | <b>Data Validation</b> | <b>Statistical Reasoning</b> | <b>Documentation</b> |
|--------------------|--------------|------------------------|------------------------------|----------------------|

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